

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

WPS301

Is your AWS GovCloud (US) architecture resilient?

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AWS Resilience Hub
AWS

Shawn Asfeld

Sr. Specialist Solutions Architect
AWS Government Regions
AWS



Continuous resilience roadmap



High availability



Disaster recovery



Continuous resilience

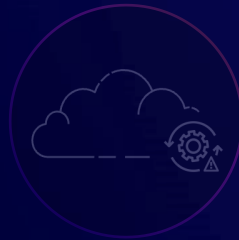
Continuous resilience roadmap



High availability

Best practices

AWS Well-Architected
Framework



Disaster recovery



Continuous resilience

Continuous resilience roadmap



High availability



Disaster recovery

**Resilience
modeling**

AWS Resilience Hub

Chaos engineering



Continuous resilience

Continuous resilience roadmap



High availability



Disaster recovery

Recovery planning

Disaster recovery

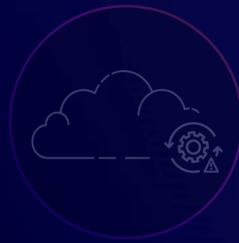


Continuous resilience

Continuous resilience roadmap



High availability



Disaster recovery



Continuous resilience

Incident analysis

Correction of Errors

re·li·abil·i·ty

A reliable workload performs its intended function correctly and consistently, when it's expected to

re·sil·ien·cy

The ability of your workload to withstand partial and intermittent failures across components

HA or DR or both

High availability

Resistance to common failures through design and operational mechanisms

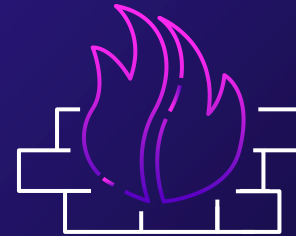


Core services, design goals to meet the availability 9s



Disaster recovery

Returning to operations within specific targets for rare but high-impact failures



Backup and recovery, data bunkering, managed RPO/RTO



Recovery spectrum

Recovery objectives

How much data can you afford
to recreate or lose?

How quickly must you recover?
What is the cost of downtime?



Strategies for disaster recovery



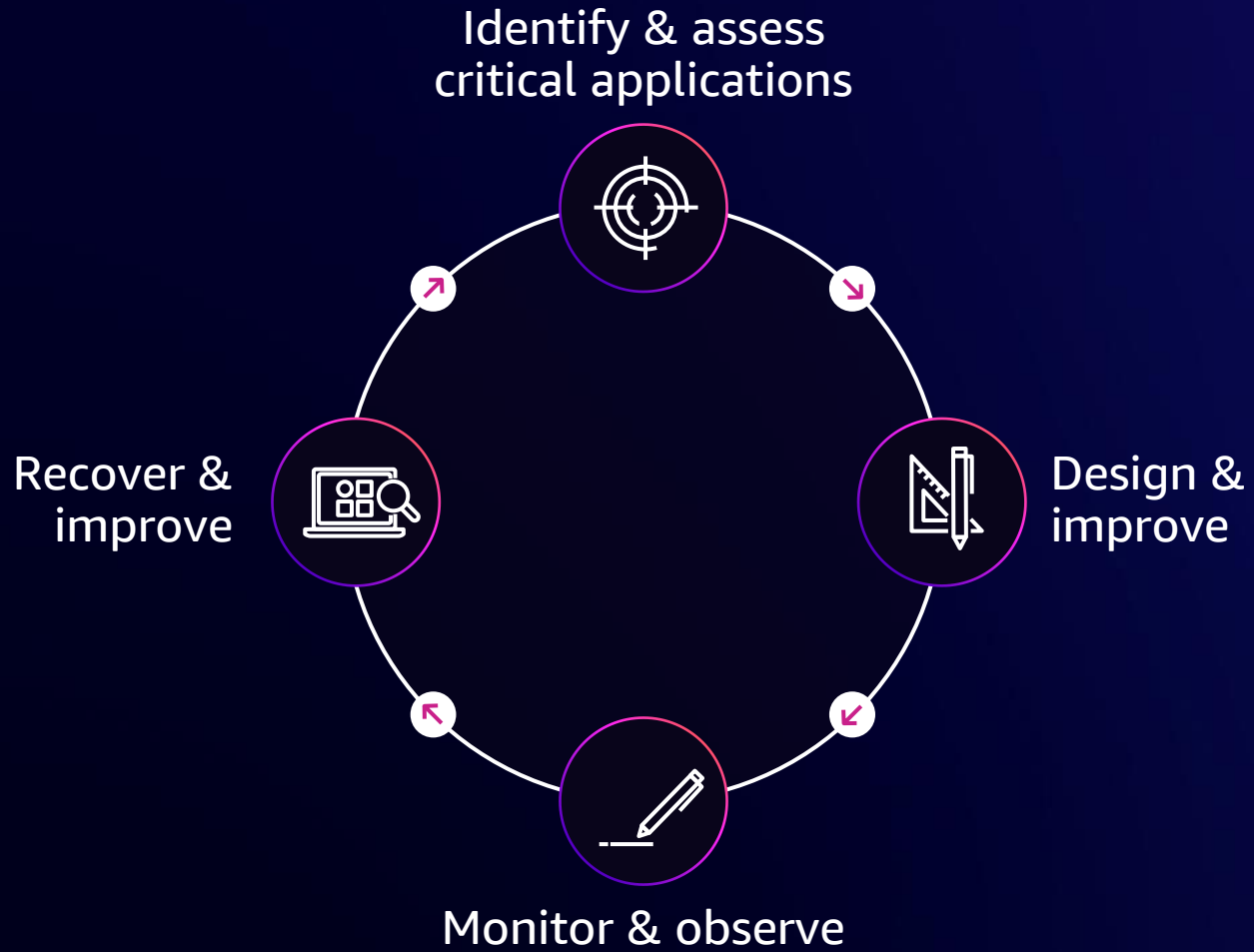


AWS is responsible for
the resilience
of the Cloud

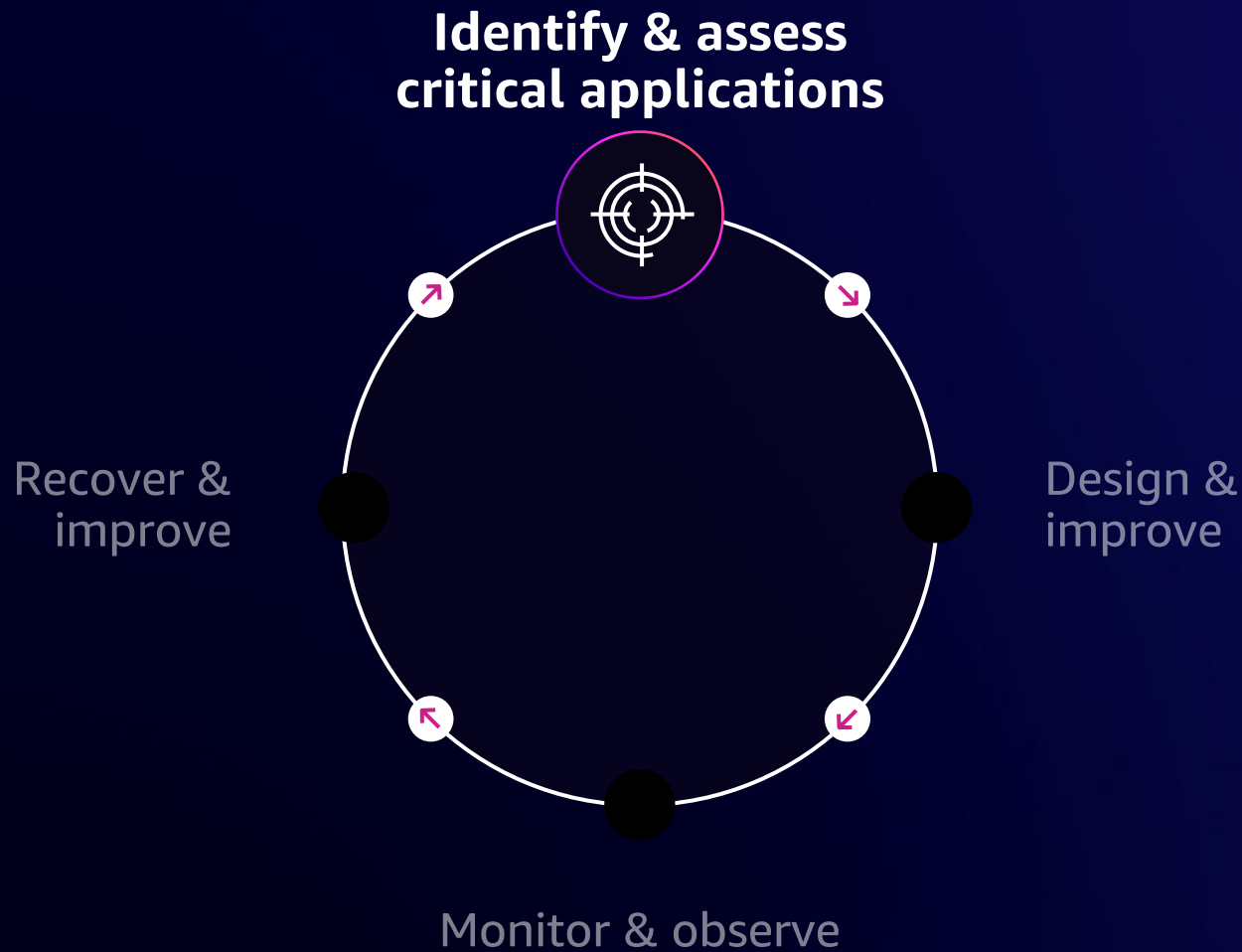


Customers are responsible for
their resilience
in the Cloud

Customer challenge



Assessing the resilience of applications



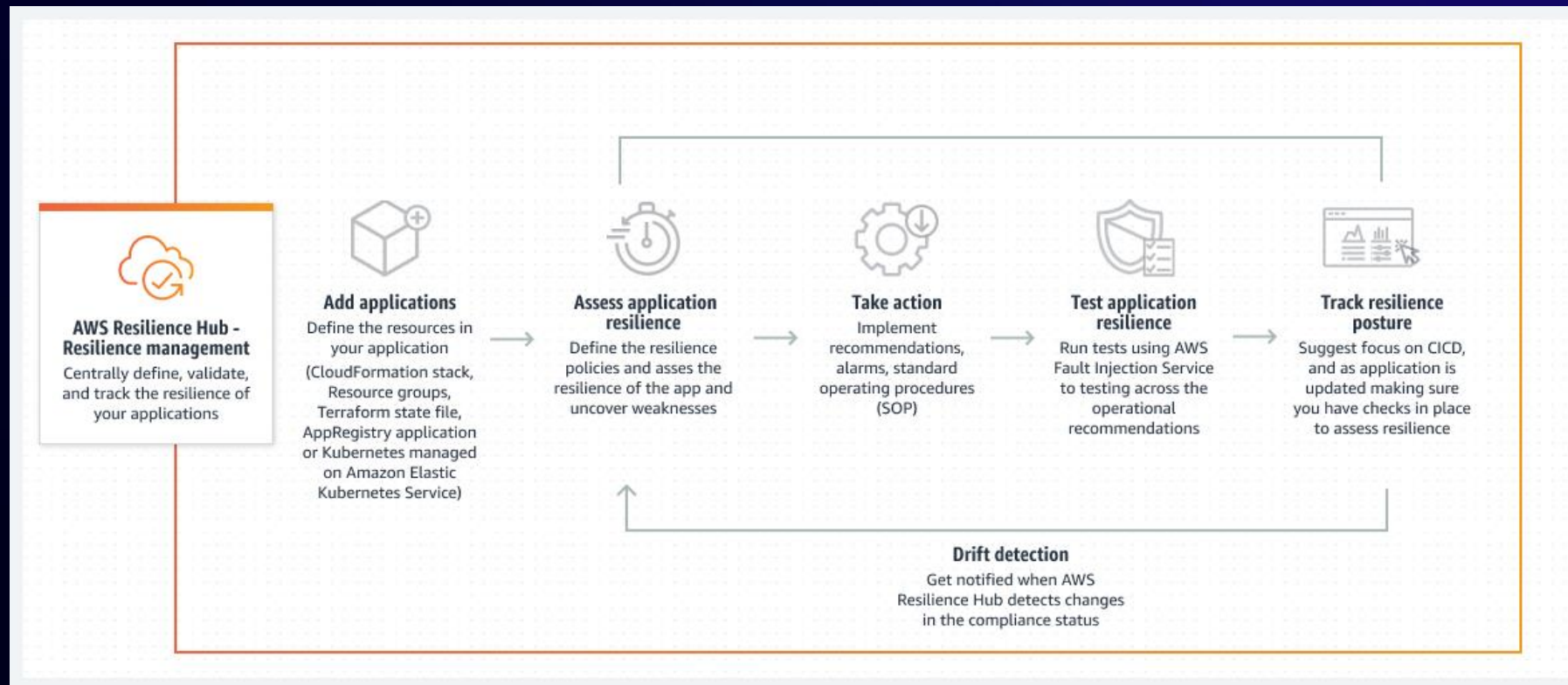
**AWS
Resilience
Hub**

What is AWS Resilience Hub?

An application resilience service that provides customers a central place to define, evaluate, validate, and track the resilience of their applications on AWS



AWS Resilience Hub



Resilience Hub

KEY CAPABILITIES

Assess



Assess whether or not
meet RTO/RPO

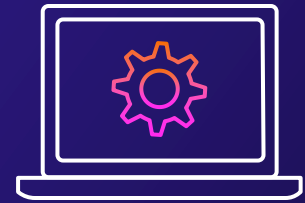
Can schedule daily
Or run as part of CI/CD

Resilience recommendations



Optimize for...
Availability Zone RTO/RPO
Cost
Minimal changes

Operational recommendations



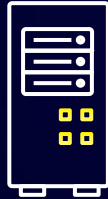
Alarms
Standard operating
procedures (SOP)
Fault injection experiments

Disruption types



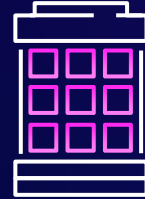
Application data

- Data deletion
- Data corruption
- Software bugs



Infrastructure

- Component failure:
 - Hard drive
 - Power supply
- In AWS may look like:
 - Amazon EC2 instance failure
 - Amazon RDS instance failure



Availability Zone

- Natural disaster:
 - Fire
 - Flood
- Technical failure
 - Power outage
 - Network outage



Region

- Unusual natural disaster
 - Act of God
- Technical failure
 - Region-wide outage

Using AWS Resilience Hub



Resilience Hub users



Define resilience targets

Using AWS Resilience Hub



Resilience Hub users



Define
resilience
targets



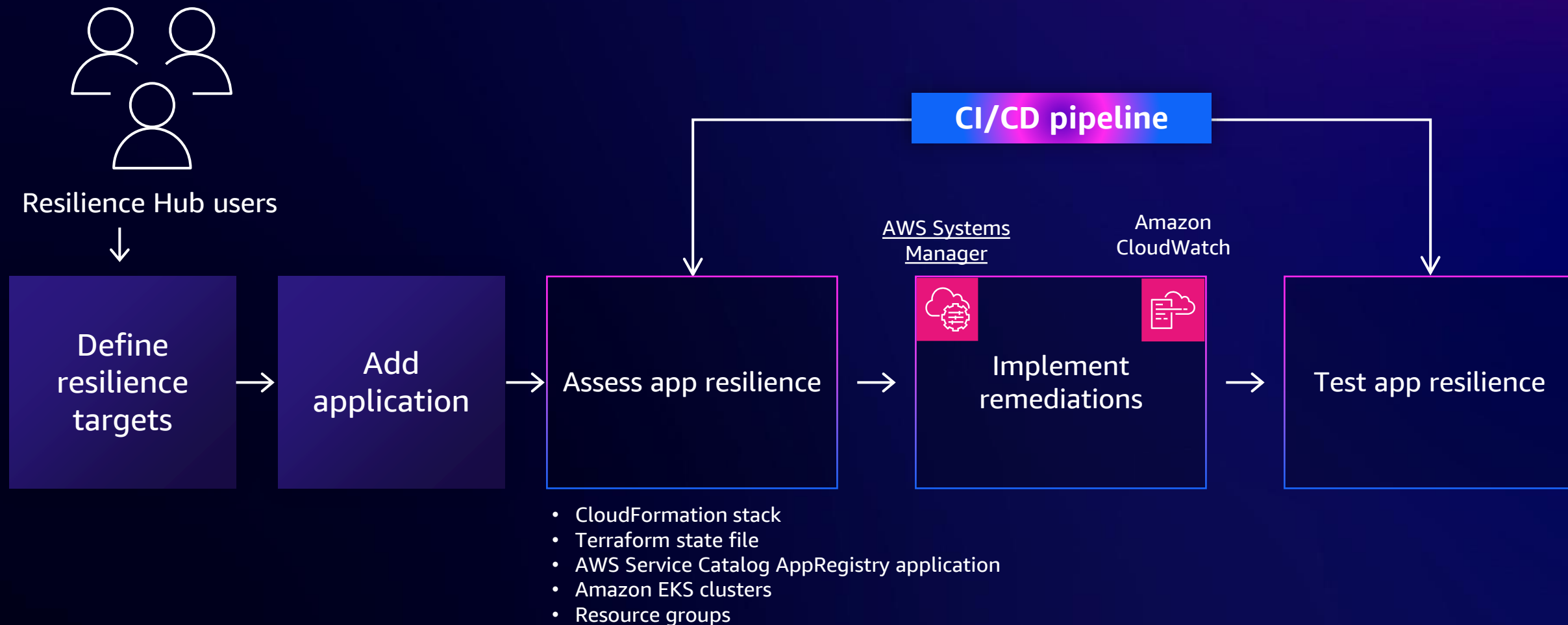
Add application

IaC tools

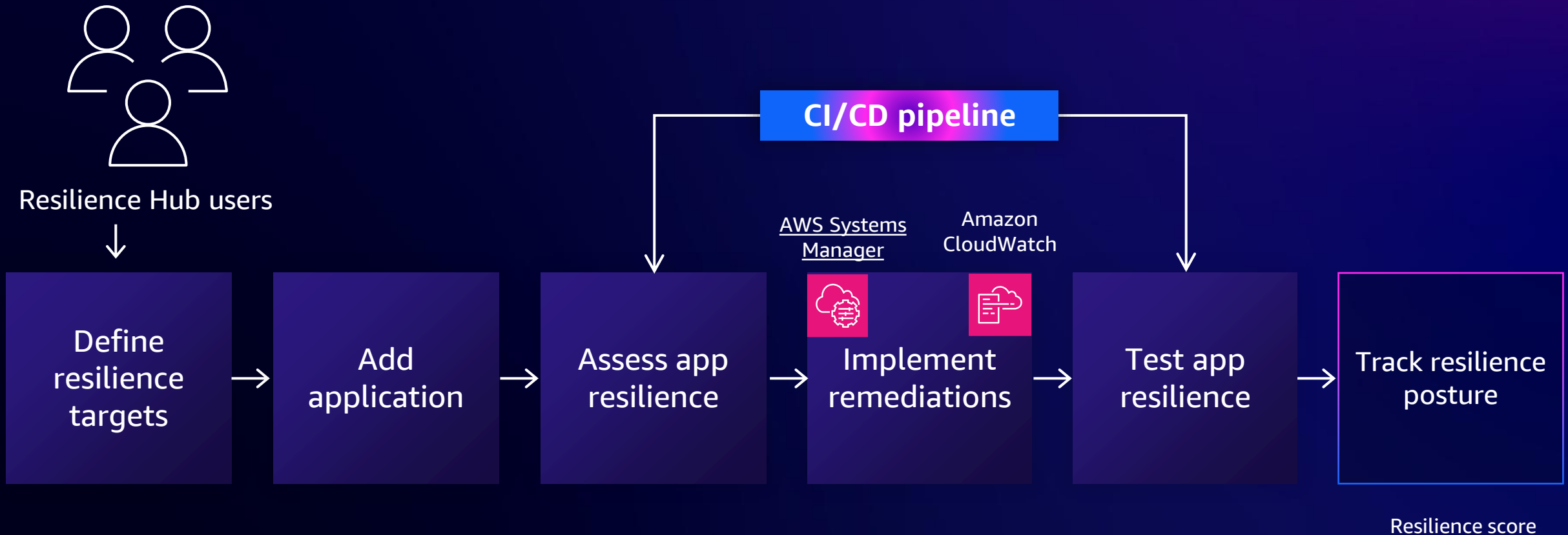
- CloudFormation stack
- Terraform state file

- AWS Service Catalog AppRegistry
- Amazon EKS clusters
- Resource groups

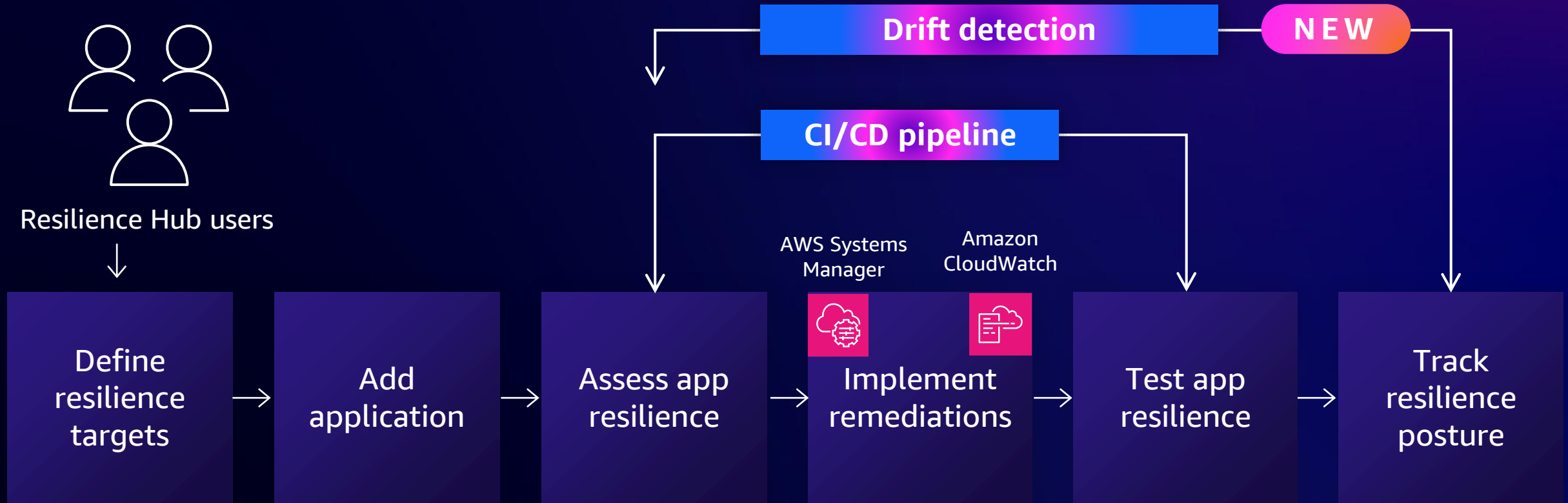
Using AWS Resilience Hub



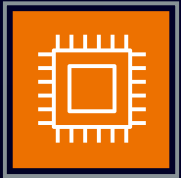
Using AWS Resilience Hub



Using AWS Resilience Hub



Supported resources



Compute

Amazon EC2
Amazon EKS
Amazon ECS
AWS Lambda
AWS Auto Scaling
AWS Step Functions



Networking

NAT gateway
Amazon Route 53
Elastic Load Balancing (ELB)
Amazon Route 53 Application Recovery Controller



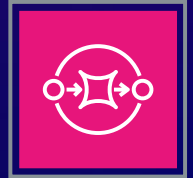
Database

Amazon RDS
Amazon Aurora
Amazon DynamoDB
Amazon DocumentDB



Storage

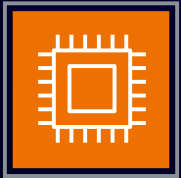
Amazon S3
Amazon EBS
Amazon EFS



Other

Amazon SQS
Amazon SNS
AWS Backup
Amazon API Gateway
AWS Elastic Disaster Recovery

Supported resources



Compute

Amazon EC2

Amazon EKS

Amazon ECS

AWS Lambda

AWS Auto Scaling

AWS Step Functions



Networking

NAT gateway

Amazon Route 53

Elastic Load Balancing (ELB)

Amazon Route 53
Application Recovery
Controller



Database

Amazon RDS

Amazon Aurora

Amazon
DynamoDB

Amazon
DocumentDB

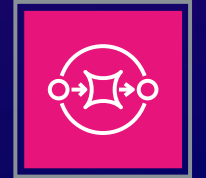


Storage

Amazon S3

Amazon EBS

Amazon EFS



Other

Amazon SQS

Amazon SNS

AWS Backup

Amazon API Gateway

AWS Elastic Disaster
Recovery

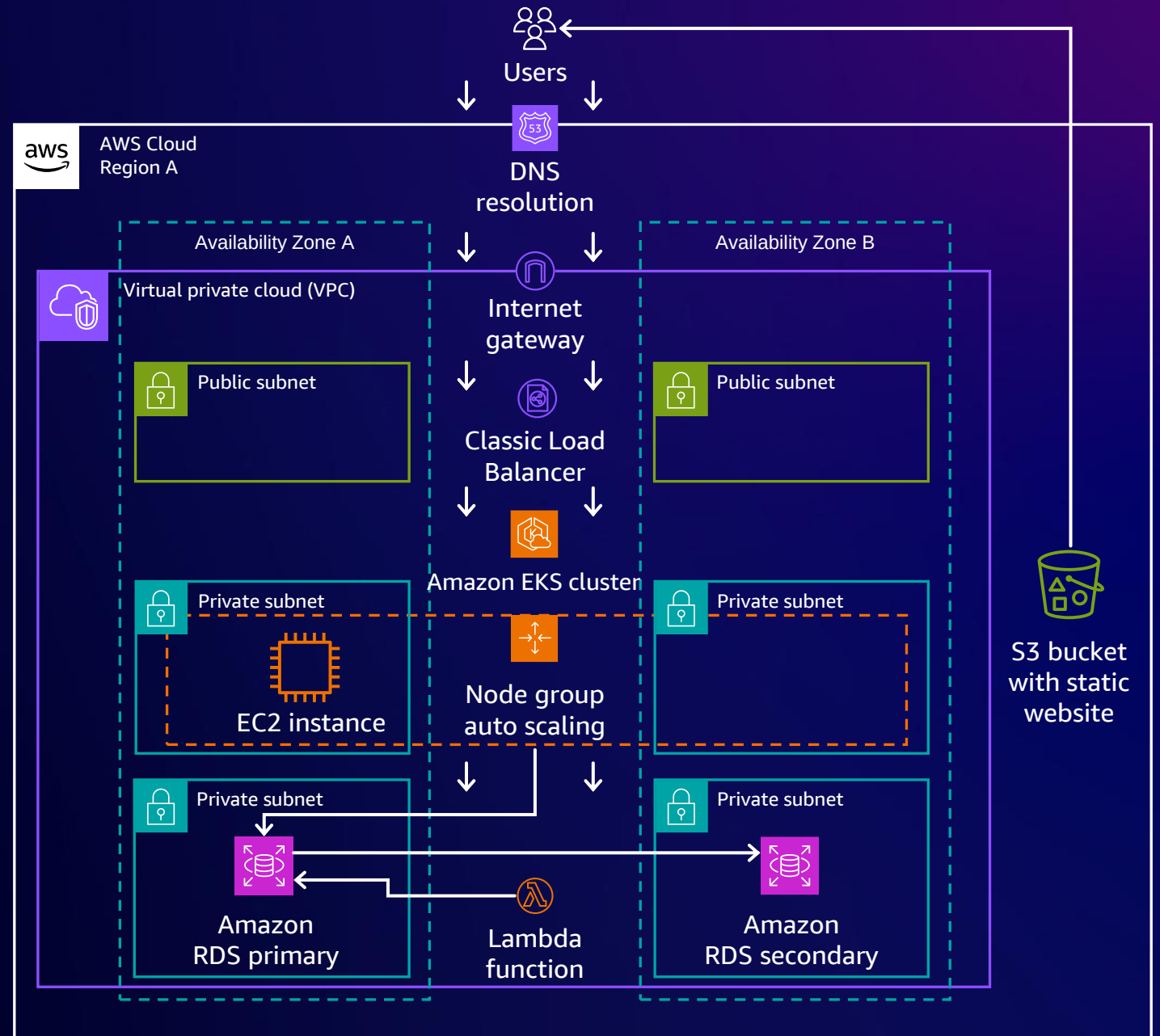
Workshop time

LET'S GET HANDS ON



E-commerce website

- Static website deployed to S3 bucket
- Application server based on Amazon EKS cluster
- REST APIs to the public internet through Classic Load Balancer
- Lambda function called by Amazon EKS cluster
- Amazon RDS Multi-AZ instance with primary writer instance and secondary instance used in case of failover



Getting started with this workshop



You have access to an AWS account with any optional pre-provisioned infrastructure and IAM policies needed to complete this workshop.



The AWS account is only available for the duration of this workshop.
You will lose access to the account once the workshop is complete.

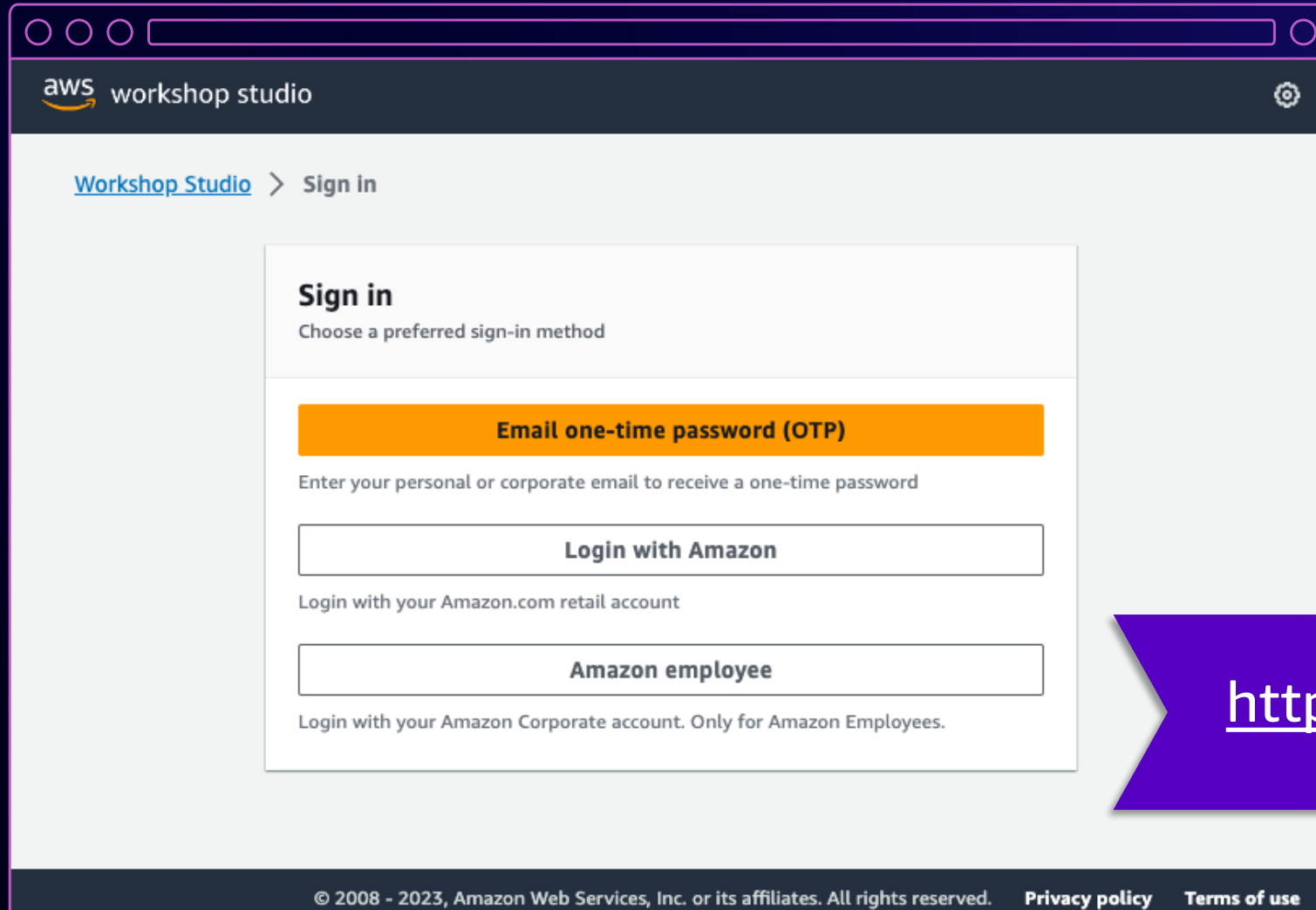


Any optional pre-provisioned infrastructure is deployed to a specific AWS Region. Make sure that you are working in this Region; other Regions are blocked.



Review the terms and conditions of the event. **Do not upload any personal or confidential information to the account.**

Step 1: Sign in using your preferred method



The screenshot shows a web browser window with the address bar empty. The page header is dark gray with the AWS logo and 'workshop studio' text on the left, and a settings gear icon on the right. Below the header, a breadcrumb trail shows 'Workshop Studio' followed by a right arrow and 'Sign in'. The main content area is white and contains a 'Sign in' section with the subtitle 'Choose a preferred sign-in method'. There are three sign-in options, each in a white box with a thin gray border: 1. 'Email one-time password (OTP)' with an orange header bar and the instruction 'Enter your personal or corporate email to receive a one-time password'. 2. 'Login with Amazon' with the instruction 'Login with your Amazon.com retail account'. 3. 'Amazon employee' with the instruction 'Login with your Amazon Corporate account. Only for Amazon Employees.' The footer is dark gray and contains the copyright notice '© 2008 - 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.' followed by links for 'Privacy policy' and 'Terms of use'.

aws workshop studio

[Workshop Studio](#) > Sign in

Sign in

Choose a preferred sign-in method

Email one-time password (OTP)

Enter your personal or corporate email to receive a one-time password

Login with Amazon

Login with your Amazon.com retail account

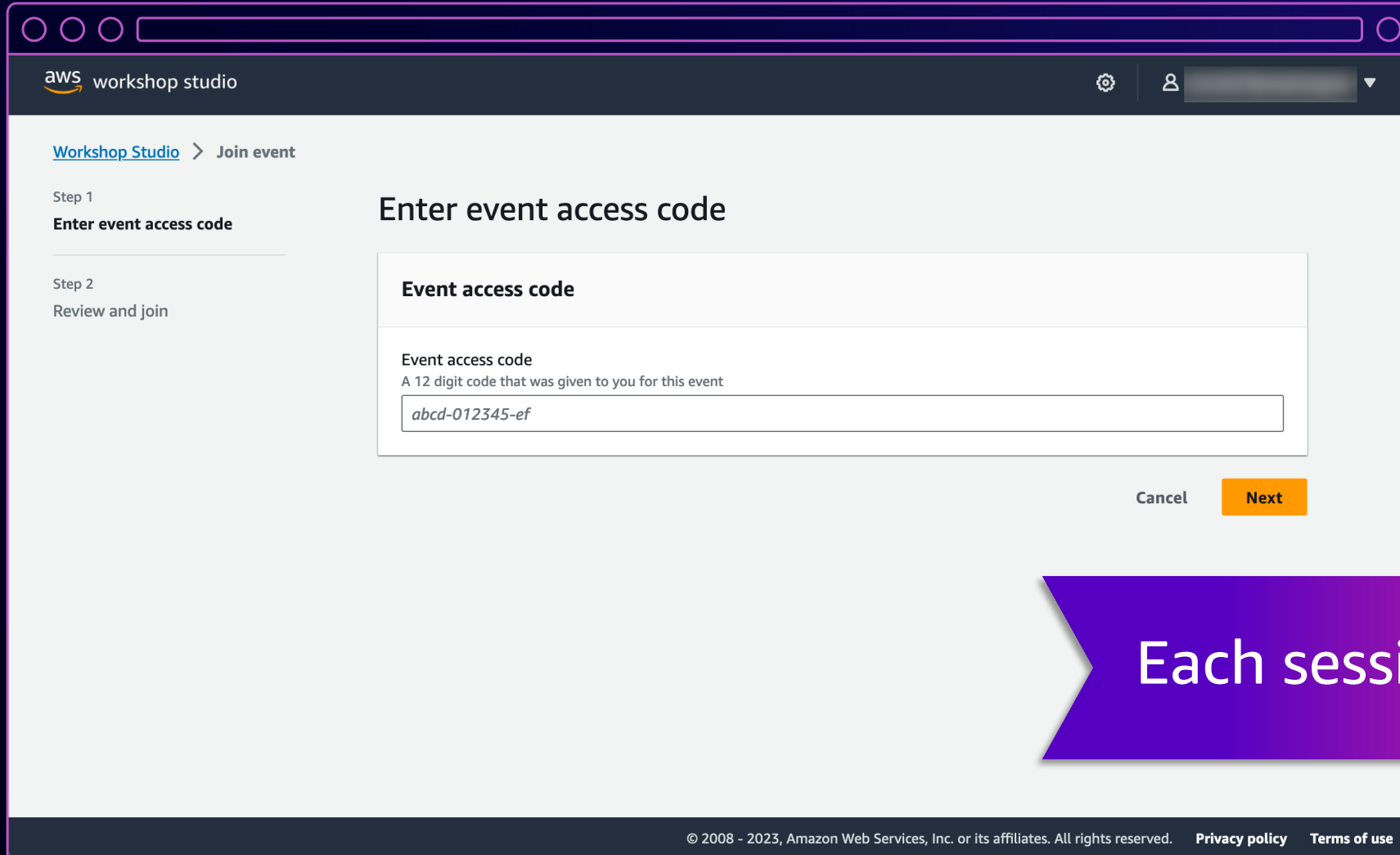
Amazon employee

Login with your Amazon Corporate account. Only for Amazon Employees.

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<https://catalog.workshops.aws/join>

Step 2: Enter the event access code



The screenshot shows the AWS Workshop Studio interface. At the top, there's a navigation bar with the AWS logo, 'workshop studio', a settings gear icon, a user profile icon, and a dropdown menu. Below the navigation bar, the breadcrumb 'Workshop Studio > Join event' is visible. On the left, a sidebar shows 'Step 1: Enter event access code' (highlighted) and 'Step 2: Review and join'. The main content area is titled 'Enter event access code'. It contains a form with the label 'Event access code' and a description: 'Event access code' followed by 'A 12 digit code that was given to you for this event'. The input field contains the text 'abcd-012345-ef'. At the bottom right of the form, there are two buttons: 'Cancel' and 'Next' (which is highlighted in orange). The footer of the page contains copyright information: '© 2008 - 2023, Amazon Web Services, Inc. or its affiliates. All rights reserved.' and links to 'Privacy policy' and 'Terms of use'.

aws workshop studio

Workshop Studio > Join event

Step 1
Enter event access code

Step 2
Review and join

Enter event access code

Event access code

Event access code
A 12 digit code that was given to you for this event

abcd-012345-ef

Cancel Next

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Each session has a unique code

Step 3: Review terms and join event

aws workshop studio

Step 1
[Enter event access code](#)

Step 2
Review and join

Review and join

Event details

Name	Start time	Duration	Level
reInvent 2023 Workshop	8/28/2023 10:59 AM	12 hours	100

Description

Workshop at re-Invent 2023

Terms and Conditions

Read and accept before joining the event

Read and accept before joining the event:

1. By using AWS Workshop Studio for the relevant event, You agree to [the AWS Event Terms and Conditions](#) and the [AWS Acceptable Use Policy](#).

2. If You are under 18 years old, You may participate in the relevant event, using AWS Workshop Studio: (a) if you are at least the minimum age below based on the country or region in which you reside, and (b) with the involvement of a parent, guardian or educator.

Country or region	Minimum age
All countries or regions not listed below (including United States, Brazil, UK, and India)	13
China, Republic of Korea (South Korea)	14
Australia	15
Japan, Switzerland, Vietnam, and countries in the European Economic Area	16
Azerbaijan, Bolivia, Colombia, Dominican Republic, Indonesia, Lebanon, Malaysia, Mexico, Montenegro, Nepal, Philippines, Thailand, Turkey, and countries in Africa	18

3. You acknowledge and agree that You are using an AWS-owned account that You will be only be able to access during the relevant event. You have no ownership rights over this AWS-owned account.

4. During the relevant event, while using this AWS-owned account, You will not use, import, input, or introduce any data, dataset, or other material that contains personal data, financial information, or any other data or materials that may be subject to laws and regulations (such as the General Data Protection Regulation or The Health Insurance Portability and Accountability Act of 1996).

5. If You find residual resources or materials in this AWS-owned account, You will notify Your Event Operator immediately.

6. AWS, its affiliates and any entities or persons acting on AWS's behalf, reserves the right to terminate this AWS-owned account and to delete its contents at any time, without any notice to You.

7. During the relevant event, while using this AWS-owned account, You will not: process or run any operation on any data other than test datasets or lab materials that have been approved by AWS.

8. You will not copy, import, export or otherwise create derivative works of materials provided by AWS for use outside of the event.

9. AWS, its affiliates and any entities or persons acting on AWS's behalf, has no obligation to enable the transmission of Your materials through AWS Workshop Studio, and may, in its discretion, edit, block, refuse to post, or remove Your materials at any time, without notice to You.

10. If You are an AWS Partner, using AWS Workshop Studio as part of Your participation in the AWS Partner Network Program, Your use of AWS Workshop Studio is governed by these terms, the AWS Partner Network Terms and Conditions, and Your applicable customer agreement with AWS.

11. Your use of AWS Workshop Studio will comply with these terms and all applicable laws. If You fail to comply with any of these terms, Your access to AWS Workshop Studio may be immediately terminated, without notice to You.

☒ I agree with the Terms and Conditions

Cancel

Previous

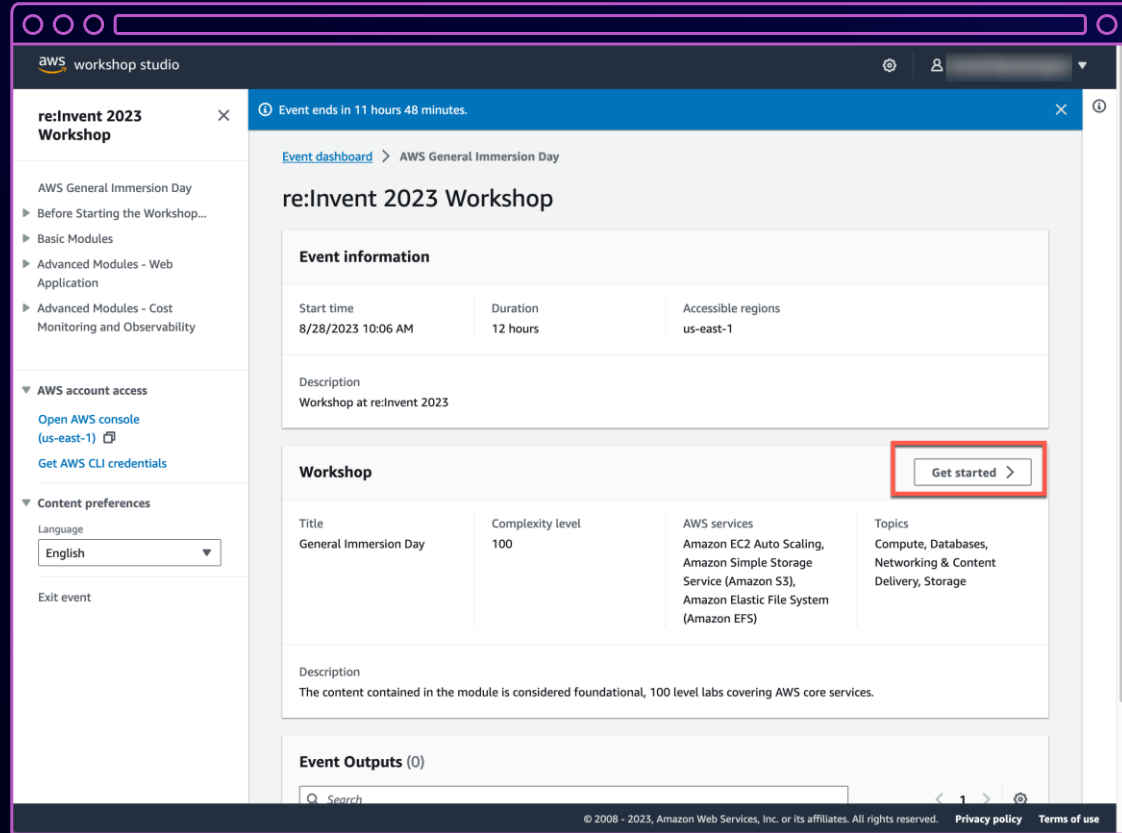
Join event

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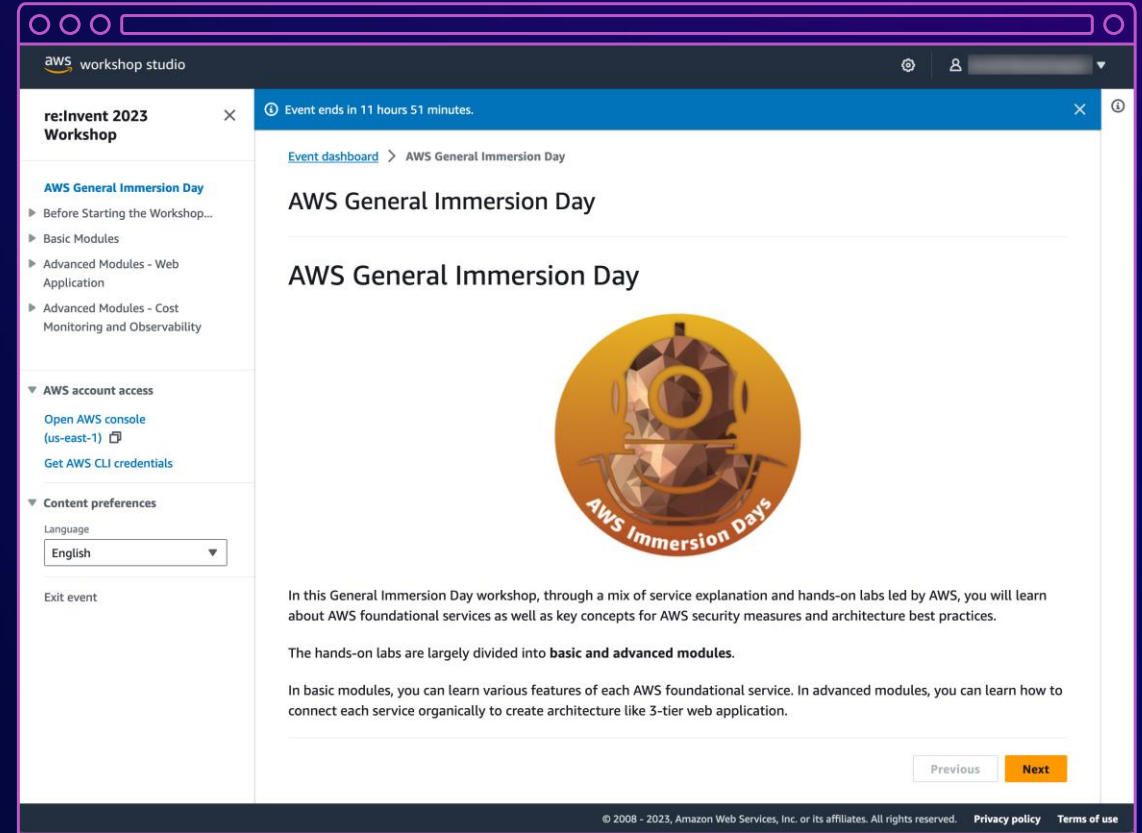
[Privacy policy](#)

[Terms of use](#)

Step 4: Get started with the workshop

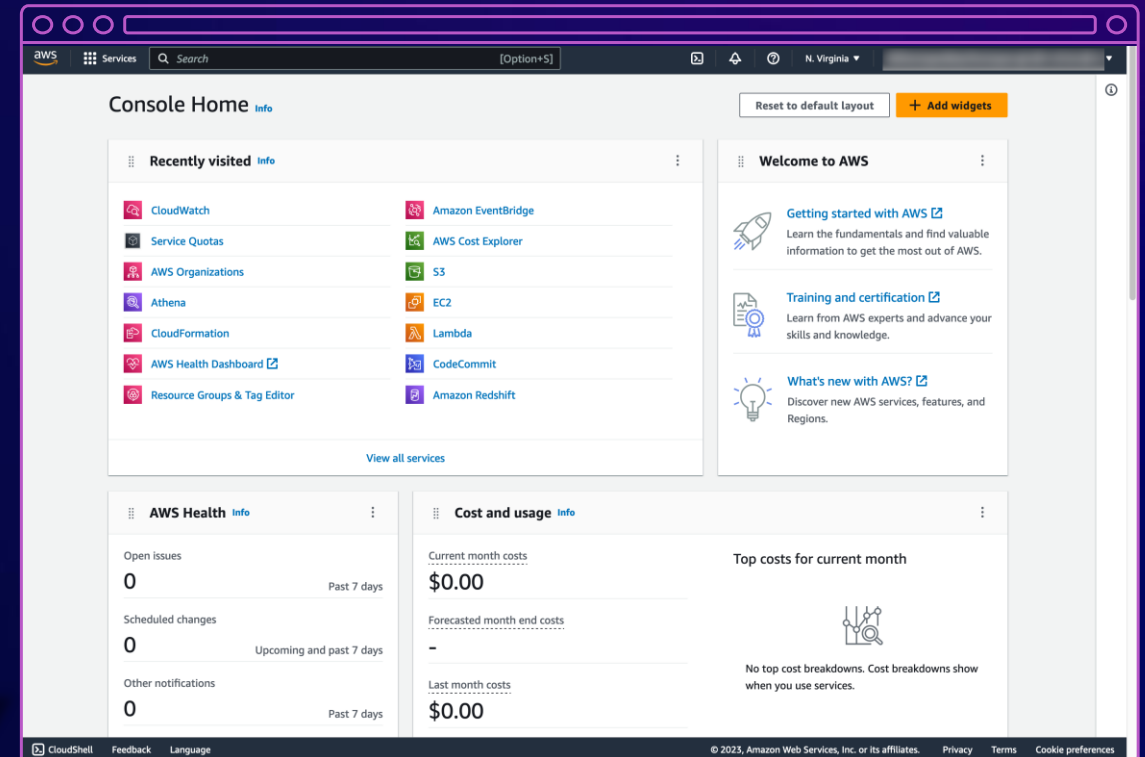
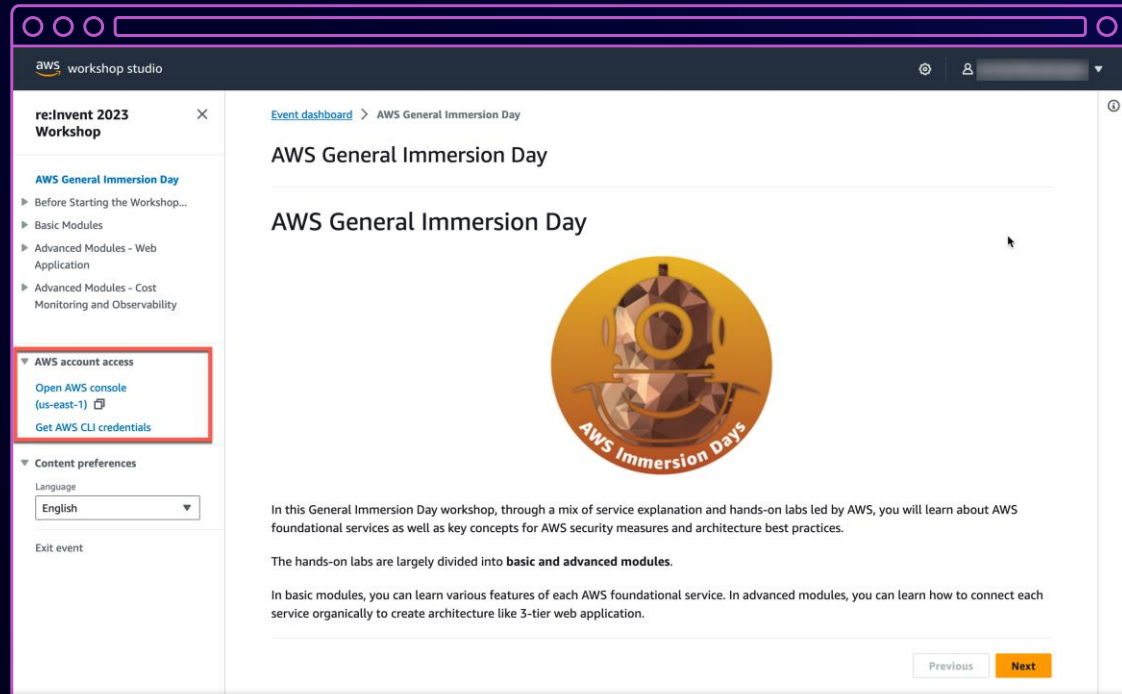


The screenshot shows the AWS Workshop Studio interface for the 're:Invent 2023 Workshop'. The left sidebar contains navigation links for 'AWS General Immersion Day', 'Before Starting the Workshop...', 'Basic Modules', 'Advanced Modules - Web Application', 'Advanced Modules - Cost Monitoring and Observability', 'AWS account access', 'Content preferences', and 'Exit event'. The main content area displays the 're:Invent 2023 Workshop' details, including event information (Start time: 8/28/2023 10:06 AM, Duration: 12 hours, Accessible regions: us-east-1), a description, and a 'Workshop' section with a 'Get started' button highlighted by a red box. The footer includes copyright information and links to Privacy policy and Terms of use.



The screenshot shows the AWS Workshop Studio interface for the 'AWS General Immersion Day' workshop. The left sidebar contains navigation links for 'AWS General Immersion Day', 'Before Starting the Workshop...', 'Basic Modules', 'Advanced Modules - Web Application', 'Advanced Modules - Cost Monitoring and Observability', 'AWS account access', 'Content preferences', and 'Exit event'. The main content area displays the 'AWS General Immersion Day' title, a large circular logo, and introductory text about the workshop content. The footer includes copyright information and links to Privacy policy and Terms of use.

Step 5: Access AWS account



Access the AWS Management Console or generate AWS CLI credentials as needed

Event access code

38dc-056734-d7

Post lab



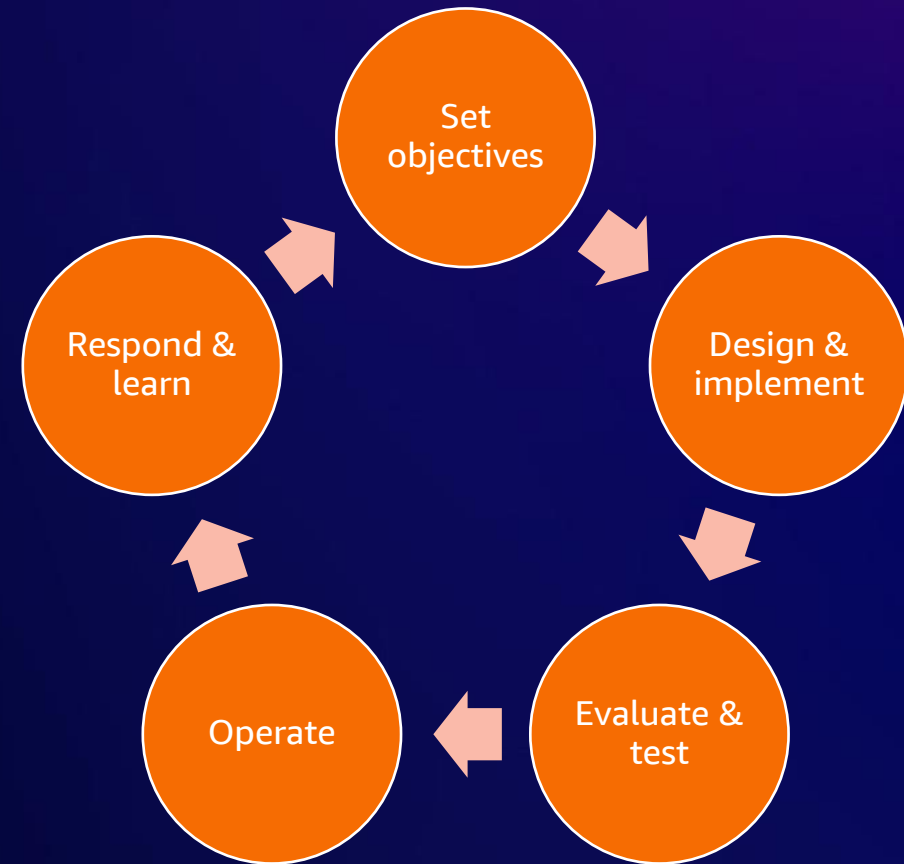
New! The AWS resilience lifecycle framework

The **resilience lifecycle framework** shares strategies, services, and mechanisms you can use to help improve your resilience posture

- Continuous process; not a one-time effort
- Modeled after a standard SDLC to easily incorporate into your existing processes



Available in the AWS Prescriptive Guidance library and aws.amazon.com/resilience



Purpose-built AWS resilience offerings

BUILD RESILIENT, HIGHLY AVAILABLE APPLICATIONS IN THE AWS CLOUD

AWS Resilience Hub

Analyze the components of your application to uncover potential resilience weaknesses

AWS Fault Injection Service (AWS FIS)

Improve application performance, observability, and resilience through controlled fault injection experiments

AWS Elastic Disaster Recovery

Minimize downtime and data loss with fast, reliable recovery of on-premises and cloud-based applications

AWS Backup

Protect data at scale using this cost-effective, fully managed, policy-based service

AWS Solutions Library

Find purpose-built AWS resilience solutions, Partner solutions, and guidance in the AWS Solutions Library



Resources



AWS Resilience Hub:

[Click here](#)

- What's new
- Training
- Videos
- Blog posts



AWS Well-Architected Reliability Pillar:

[Click here](#)

- Design principles
- Questions
- Best practices

Thank you!

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Please complete the session
survey in the mobile app

